

Marketing Analytics Regression

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Contents

1	Business Problem	2
2	Data Understanding	2
2.1	Load Libraries	2
2.2	Import Dataset	2
2.3	Initial Data Exploration	2
3	Exploratory Data Analysis	4
3.1	Correlation Analysis	4
3.2	Relationship Between PurchaseAmount and Candidate Predictors	6
3.3	Summary of Exploratory Findings	11
4	Model Development	11
4.1	Baseline Model	12
4.2	Multiple Regression Model	13
4.3	Refined Regression Model	14
4.4	Model Comparison	15
5	Model Diagnostics	16
5.1	Linearity	16
5.2	Normality of Residuals	18
5.3	Homoscedasticity	20
5.4	Multicollinearity	21
5.5	Influential Observations (Cook's Distance)	22
5.6	Diagnostic Summary	23
6	Conclusion	23

```
knitr::opts_chunk$set(
  echo = TRUE,
  warning = FALSE,
  message = FALSE
)
```

1 Business Problem

The objective of this project is to identify the factors that influence customer purchase amounts using multiple linear regression. Understanding these relationships can help organizations improve marketing strategies, optimize promotional efforts, and better understand customer purchasing behavior.

2 Data Understanding

2.1 Load Libraries

Load the packages required for data import, visualization, statistical analysis, and regression modeling.

```
library(tidyverse)
library(readxl)
library(corrplot)
library(car)
library(lmtest)
library(knitr)
```

2.2 Import Dataset

Import the dataset and create the main data frame for the analysis.

```
marketing <- read_excel("../data/marketing_customer_data.xlsx")
```

2.3 Initial Data Exploration

Examine the dataset structure, variable types, descriptive statistics, and missing values to assess data quality before the exploratory analysis.

```
# Display the structure of the dataset
str(marketing)
```

```
## tibble [5,000 x 15] (S3: tbl_df/tbl/data.frame)
## $ CustomerID      : num [1:5000] 1 2 3 4 5 6 7 8 9 10 ...
## $ Age              : num [1:5000] 56 69 46 32 60 25 38 56 36 40 ...
## $ Gender           : chr [1:5000] "Male" "Male" "Female" "Female" ...
## $ Income           : num [1:5000] 48828 105115 73105 50711 70155 ...
## $ Region           : chr [1:5000] "North" "West" "East" "South" ...
## $ WebsiteVisits    : num [1:5000] 5 4 10 14 8 14 11 7 8 8 ...
## $ TimeOnSite       : num [1:5000] 4.28 7.1 5.26 1.6 2.57 ...
```

```
## $ PagesViewed      : num [1:5000] 9 9 7 2 7 9 18 12 7 19 ...
## $ AdImpressions    : num [1:5000] 137 476 280 95 451 56 122 230 289 85 ...
## $ Clicks           : num [1:5000] 2 2 2 2 3 2 7 4 2 1 ...
## $ EmailOpens       : num [1:5000] 11 24 5 10 5 19 4 12 0 15 ...
## $ PromoUsed        : num [1:5000] 0 0 0 0 0 1 1 0 1 0 ...
## $ SatisfactionScore: num [1:5000] 3.32 3.02 4.26 3.73 4.28 4.73 3.85 2.58 1.85 2.08 ...
## $ PurchaseAmount   : num [1:5000] 50 56.6 77.3 63.1 66 ...
## $ Churn            : num [1:5000] 0 0 1 0 0 0 1 0 0 0 ...
```

```
# Display a compact overview
glimpse(marketing)
```

```
## Rows: 5,000
## Columns: 15
## $ CustomerID      <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 1~
## $ Age             <dbl> 56, 69, 46, 32, 60, 25, 38, 56, 36, 40, 28, 28, 41, ~
## $ Gender          <chr> "Male", "Male", "Female", "Female", "Male", "Female"~
## $ Income          <dbl> 48828.40, 105114.81, 73104.65, 50710.55, 70155.34, 9~
## $ Region          <chr> "North", "West", "East", "South", "North", "South", ~
## $ WebsiteVisits   <dbl> 5, 4, 10, 14, 8, 14, 11, 7, 8, 8, 9, 19, 11, 4, 13, ~
## $ TimeOnSite      <dbl> 4.279425, 7.104381, 5.263880, 1.602441, 2.573624, 8.~
## $ PagesViewed     <dbl> 9, 9, 7, 2, 7, 9, 18, 12, 7, 19, 16, 8, 19, 6, 2, 2,~
## $ AdImpressions   <dbl> 137, 476, 280, 95, 451, 56, 122, 230, 289, 85, 460, ~
## $ Clicks          <dbl> 2, 2, 2, 2, 3, 2, 7, 4, 2, 1, 2, 1, 3, 4, 2, 3, 1, 1~
## $ EmailOpens      <dbl> 11, 24, 5, 10, 5, 19, 4, 12, 0, 15, 18, 10, 21, 1, 0~
## $ PromoUsed       <dbl> 0, 0, 0, 0, 0, 0, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0~
## $ SatisfactionScore <dbl> 3.32, 3.02, 4.26, 3.73, 4.28, 4.73, 3.85, 2.58, 1.85~
## $ PurchaseAmount  <dbl> 49.98990, 56.57952, 77.31257, 63.14771, 65.96856, 13~
## $ Churn           <dbl> 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 1, 1, 0, 0, 0~
```

```
# Generate descriptive statistics
summary(marketing)
```

```
##      CustomerID      Age      Gender      Income
## Min.   : 1      Min.   :18.00      Length :5000      Min.   : 30000
## 1st Qu.:1251    1st Qu.:32.00      N.unique : 2      1st Qu.: 72513
## Median :2500    Median :46.00      N.blank  : 0      Median : 89562
## Mean   :2500    Mean   :46.15      Min.nchar: 4      Mean   : 89633
## 3rd Qu.:3750    3rd Qu.:60.00      Max.nchar: 6      3rd Qu.:106737
## Max.   :5000    Max.   :74.00                      Max.   :178602
##      Region      WebsiteVisits      TimeOnSite      PagesViewed
## Length :5000      Min.   : 0.00      Min.   : 0.500      Min.   : 1.000
## N.unique : 4      1st Qu.: 8.00      1st Qu.: 3.572      1st Qu.: 5.000
## N.blank  : 0      Median :10.00      Median : 4.983      Median :10.000
## Min.nchar: 4      Mean   :10.03      Mean   : 4.995      Mean   : 9.889
## Max.nchar: 5      3rd Qu.:12.00      3rd Qu.: 6.319      3rd Qu.:15.000
## Max.   :23.00      Max.   :12.375      Max.   :19.000
##      AdImpressions      Clicks      EmailOpens      PromoUsed
## Min.   : 0.0      Min.   : 0.000      Min.   : 0.00      Min.   :0.0000
## 1st Qu.:124.0      1st Qu.: 2.000      1st Qu.: 7.00      1st Qu.:0.0000
## Median :249.0      Median : 3.000      Median :14.00      Median :0.0000
## Mean   :249.3      Mean   : 2.991      Mean   :14.49      Mean   :0.4966
## 3rd Qu.:376.0      3rd Qu.: 4.000      3rd Qu.:22.00      3rd Qu.:1.0000
```

```
## Max. :499.0 Max. :13.000 Max. :29.00 Max. :1.0000
## SatisfactionScore PurchaseAmount Churn
## Min. :1.000 Min. : 7.234 Min. :0.000
## 1st Qu.:2.018 1st Qu.: 75.236 1st Qu.:0.000
## Median :3.030 Median : 89.313 Median :0.000
## Mean :3.008 Mean : 89.706 Mean :0.244
## 3rd Qu.:4.010 3rd Qu.:104.243 3rd Qu.:0.000
## Max. :5.000 Max. :167.109 Max. :1.000
```

```
# Check for missing values
colSums(is.na(marketing))
```

```
## CustomerID Age Gender Income
## 0 0 0 0
## Region WebsiteVisits TimeOnSite PagesViewed
## 0 0 0 0
## AdImpressions Clicks EmailOpens PromoUsed
## 0 0 0 0
## SatisfactionScore PurchaseAmount Churn
## 0 0 0
```

Key Findings

The dataset contains 5,000 observations and 15 variables, with no missing values. Most variables are numerical, while Gender and Region are categorical. PurchaseAmount was selected as the response variable because it is a continuous variable and aligns with the objective of identifying the factors that influence customer spending.

3 Exploratory Data Analysis

Explore the relationships between the response variable and the candidate predictors to support the selection of variables for the regression models.

3.1 Correlation Analysis

Examine the linear relationships among the numerical variables and identify the predictors most strongly associated with PurchaseAmount.

```
# Select numeric variables
numeric_data <- marketing %>%
  select(-CustomerID) %>%
  select(where(is.numeric))

# Compute the correlation matrix
cor_matrix <- cor(numeric_data)

# Display correlation coefficients
round(cor_matrix, 2)
```

```
## Age Income WebsiteVisits TimeOnSite PagesViewed
## Age 1.00 -0.02 0.00 0.01 0.00
```

```

## Income      -0.02  1.00      0.01      0.00      0.01
## WebsiteVisits 0.00  0.01      1.00      0.01     -0.01
## TimeOnSite   0.01  0.00      0.01      1.00     -0.02
## PagesViewed  0.00  0.01     -0.01     -0.02      1.00
## AdImpressions 0.00 -0.01      0.01      0.03     -0.02
## Clicks        0.00 -0.02      0.01     -0.02      0.01
## EmailOpens    0.00  0.00      0.02      0.00     -0.01
## PromoUsed     0.02 -0.04     -0.01     -0.01     -0.02
## SatisfactionScore -0.01 -0.01     -0.01     -0.02      0.01
## PurchaseAmount 0.00  0.58      0.29      0.00      0.37
## Churn         -0.01  0.01      0.01     -0.01     -0.01
##
##              AdImpressions Clicks EmailOpens PromoUsed SatisfactionScore
## Age              0.00  0.00      0.00      0.02      -0.01
## Income           -0.01 -0.02      0.00     -0.04      -0.01
## WebsiteVisits     0.01  0.01      0.02     -0.01      -0.01
## TimeOnSite        0.03 -0.02      0.00     -0.01      -0.02
## PagesViewed      -0.02  0.01     -0.01     -0.02      0.01
## AdImpressions     1.00  0.02      0.00      0.00      0.02
## Clicks            0.02  1.00      0.01      0.00      0.02
## EmailOpens        0.00  0.01      1.00      0.02     -0.01
## PromoUsed         0.00  0.00      0.02      1.00      0.00
## SatisfactionScore 0.02  0.02     -0.01      0.00      1.00
## PurchaseAmount    -0.01 -0.01      0.00      0.44     -0.01
## Churn             0.02 -0.01      0.00      0.00      0.02
##
##              PurchaseAmount Churn
## Age              0.00 -0.01
## Income           0.58  0.01
## WebsiteVisits    0.29  0.01
## TimeOnSite       0.00 -0.01
## PagesViewed      0.37 -0.01
## AdImpressions    -0.01  0.02
## Clicks           -0.01 -0.01
## EmailOpens       0.00  0.00
## PromoUsed        0.44  0.00
## SatisfactionScore -0.01  0.02
## PurchaseAmount   1.00  0.00
## Churn            0.00  1.00

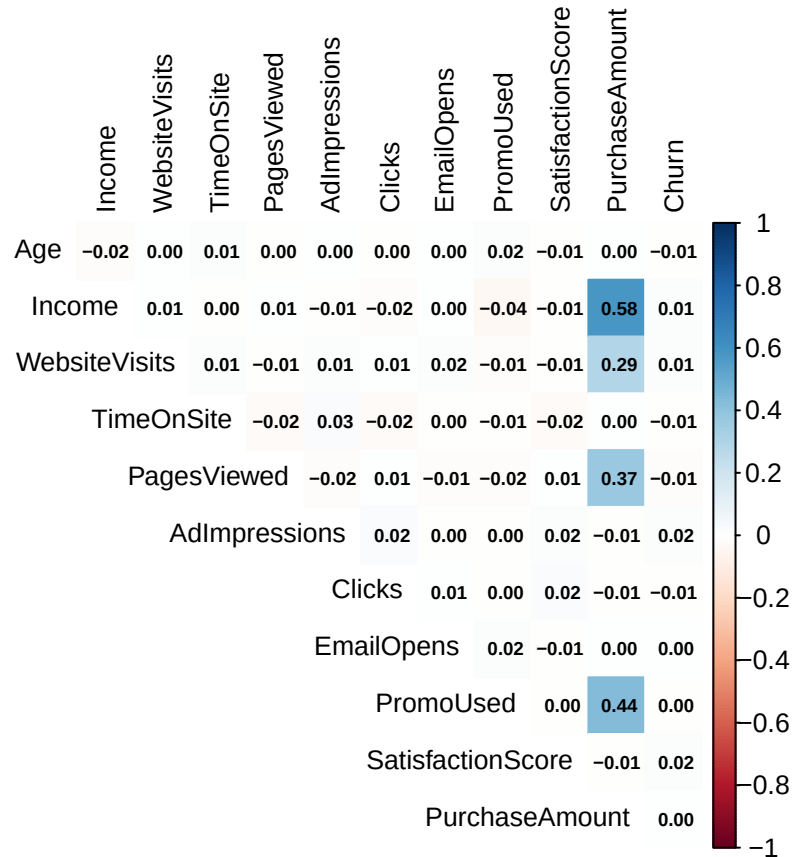
```

Visualize the correlation matrix

```

corrplot(
  cor_matrix,
  method = "color",
  type = "upper",
  addCoef.col = "black",
  tl.col = "black",
  tl.cex = 0.8,
  number.cex = 0.6,
  diag = FALSE
)

```



Key Findings

Income, PromoUsed, PagesViewed, and WebsiteVisits showed the strongest positive correlations with PurchaseAmount. Although SatisfactionScore exhibited a weak linear correlation, it was retained for further investigation due to its business relevance. The low correlations among the predictors also suggest a low risk of multicollinearity.

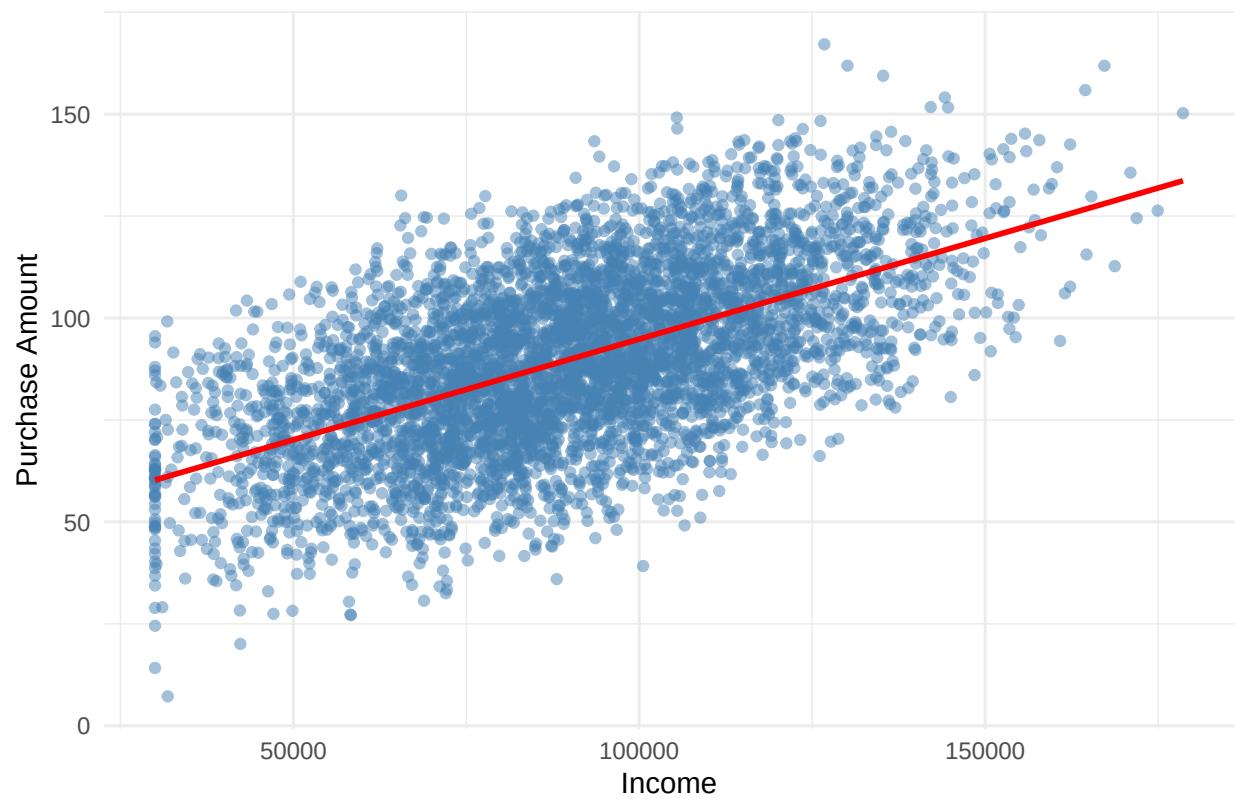
3.2 Relationship Between PurchaseAmount and Candidate Predictors

Visually examine the relationships between PurchaseAmount and the candidate predictors to assess whether they exhibit patterns consistent with linear regression.

```
# Relationship Between Income and PurchaseAmount

ggplot(marketing, aes(x = Income, y = PurchaseAmount)) +
  geom_point(alpha = 0.5, color = "steelblue") +
  geom_smooth(method = "lm", se = FALSE, color = "red") +
  labs(
    title = "Purchase Amount vs Income",
    x = "Income",
    y = "Purchase Amount"
  ) +
  theme_minimal()
```

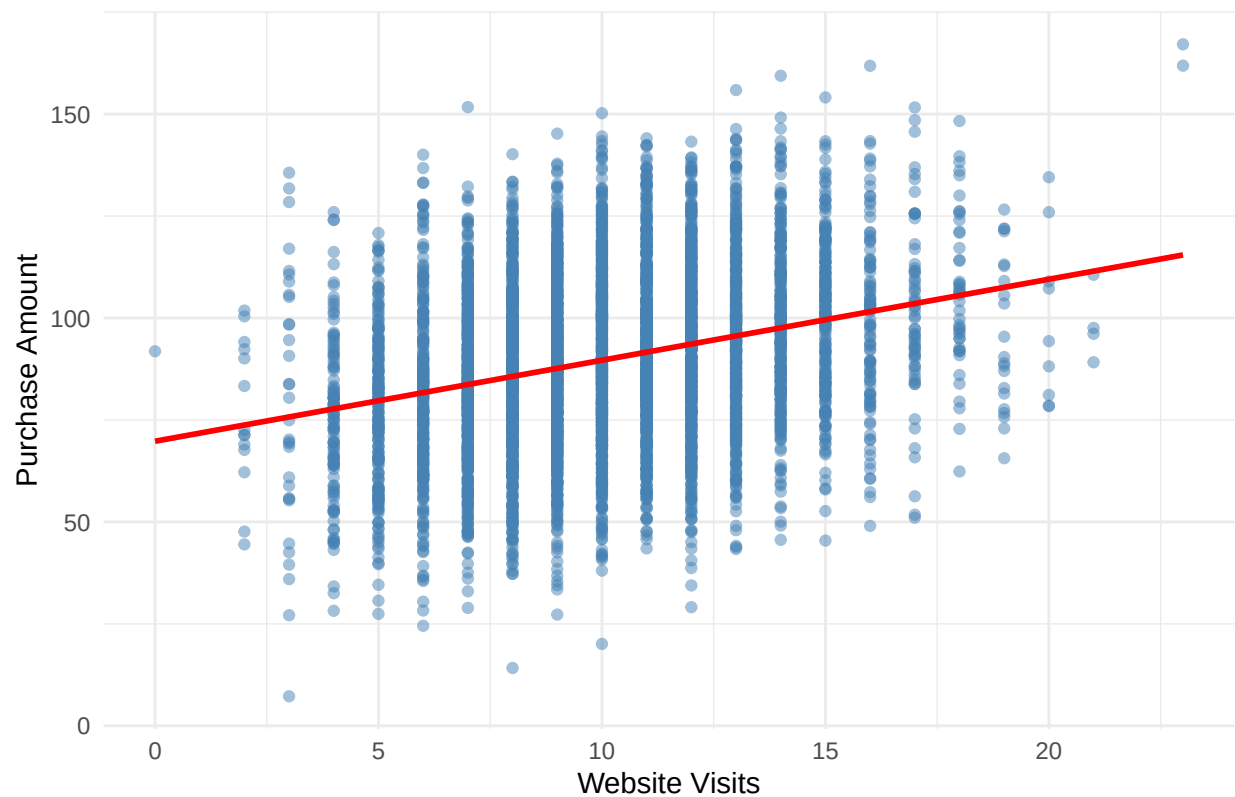
Purchase Amount vs Income



```
# Relationship Between Website Visits and PurchaseAmount

ggplot(marketing, aes(x = WebsiteVisits, y = PurchaseAmount)) +
  geom_point(alpha = 0.5, color = "steelblue") +
  geom_smooth(method = "lm", se = FALSE, color = "red") +
  labs(
    title = "Purchase Amount vs Website Visits",
    x = "Website Visits",
    y = "Purchase Amount"
  ) +
  theme_minimal()
```

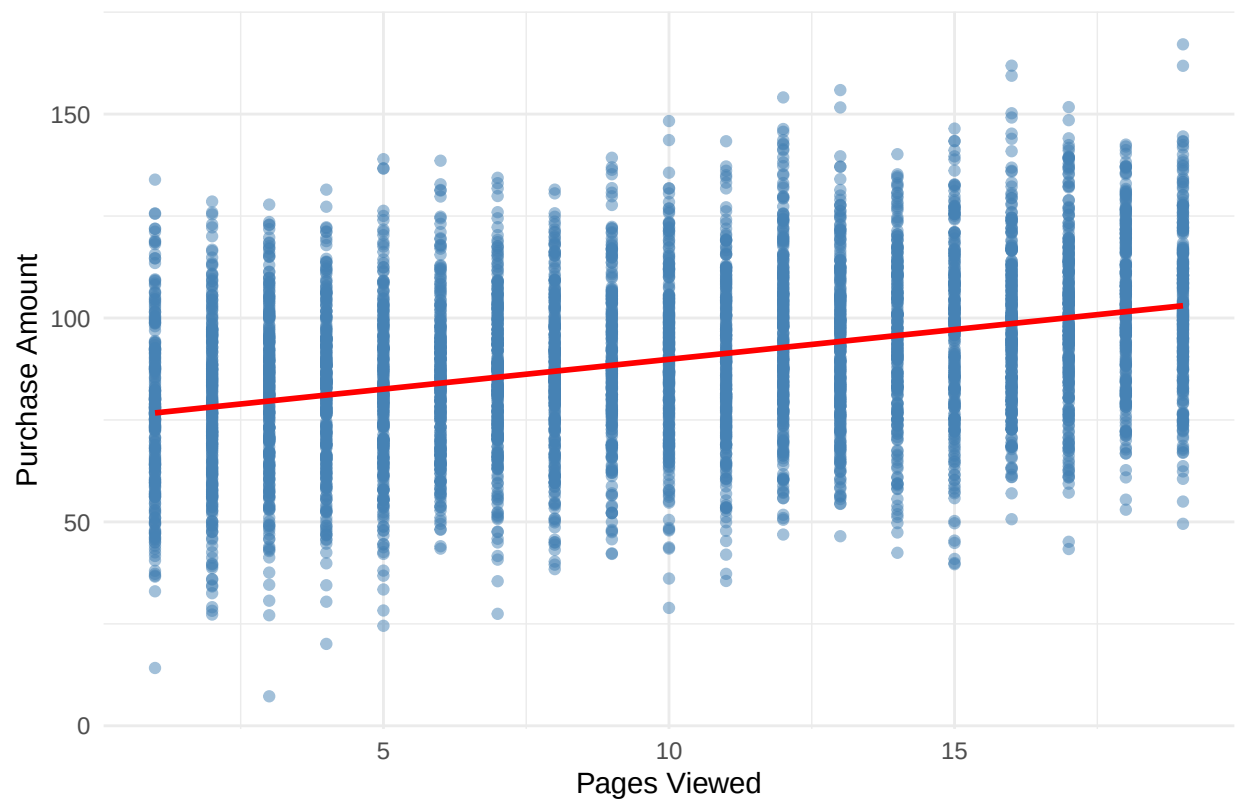
Purchase Amount vs Website Visits



```
# Relationship Between Pages Viewed and PurchaseAmount

ggplot(marketing, aes(x = PagesViewed, y = PurchaseAmount)) +
  geom_point(alpha = 0.5, color = "steelblue") +
  geom_smooth(method = "lm", se = FALSE, color = "red") +
  labs(
    title = "Purchase Amount vs Pages Viewed",
    x = "Pages Viewed",
    y = "Purchase Amount"
  ) +
  theme_minimal()
```

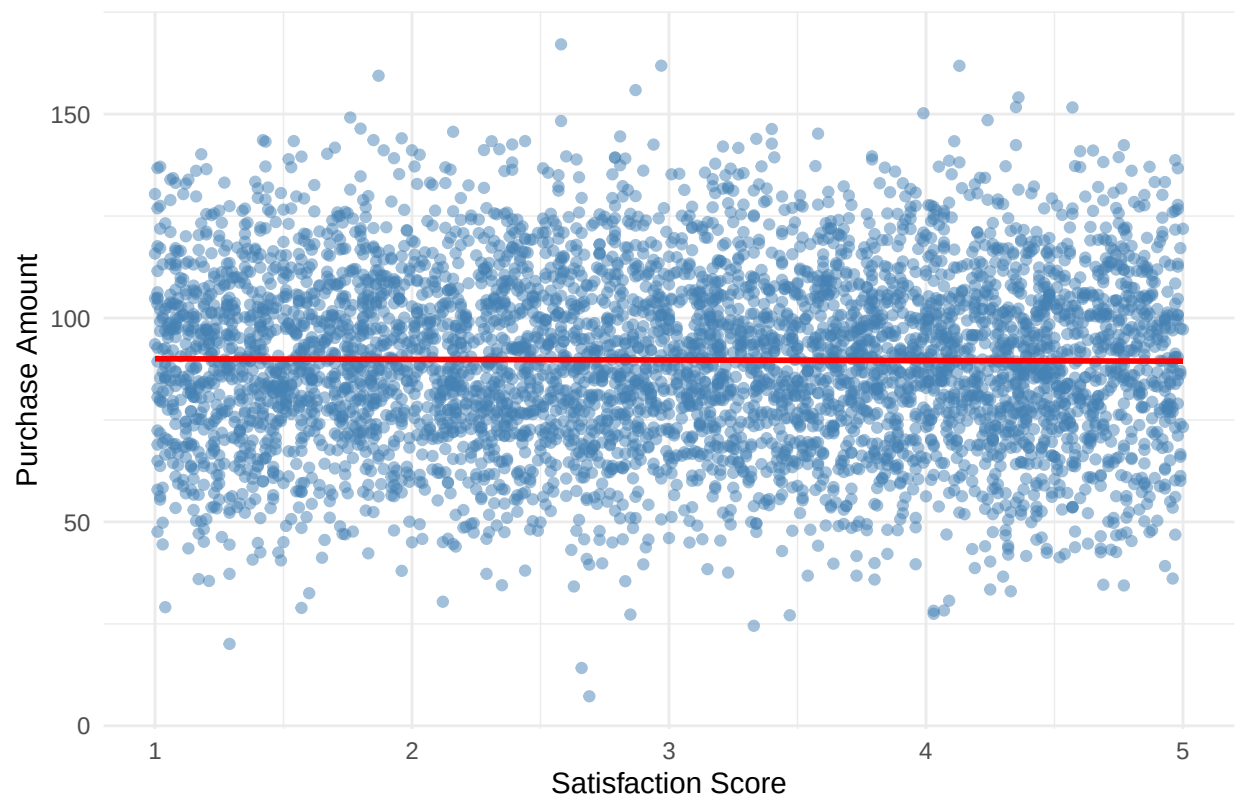

Purchase Amount vs Pages Viewed



Relationship Between Satisfaction Score and PurchaseAmount

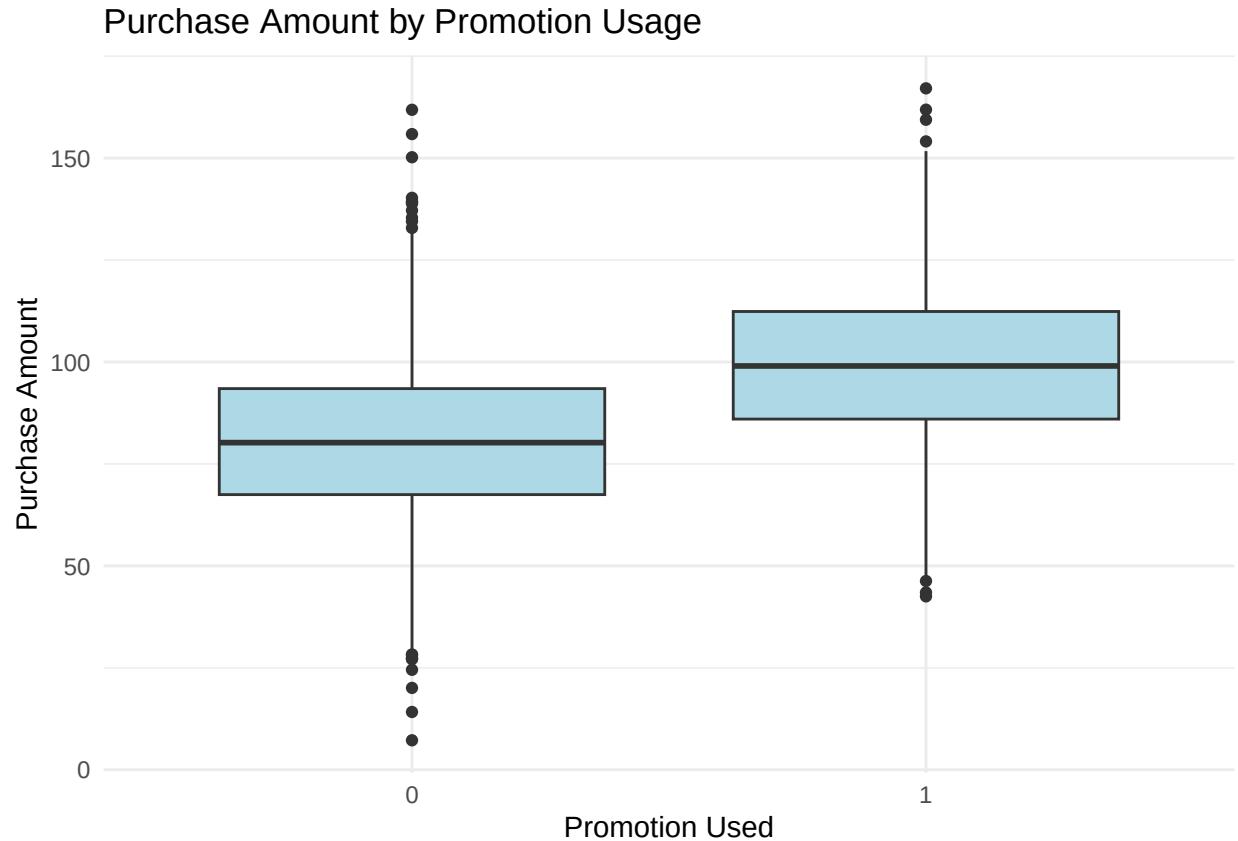
```
ggplot(marketing, aes(x = SatisfactionScore, y = PurchaseAmount)) +
  geom_point(alpha = 0.5, color = "steelblue") +
  geom_smooth(method = "lm", se = FALSE, color = "red") +
  labs(
    title = "Purchase Amount vs Satisfaction Score",
    x = "Satisfaction Score",
    y = "Purchase Amount"
  ) +
  theme_minimal()
```

Purchase Amount vs Satisfaction Score



```
# PurchaseAmount by Promotion Usage

ggplot(marketing,
  aes(x = factor(PromoUsed),
    y = PurchaseAmount)) +
  geom_boxplot(fill = "lightblue") +
  labs(
    title = "Purchase Amount by Promotion Usage",
    x = "Promotion Used",
    y = "Purchase Amount"
  ) +
  theme_minimal()
```



Key Findings

The visual analysis supported the correlation results. Income, PromoUsed, PagesViewed, and WebsiteVisits exhibited positive relationships with PurchaseAmount, although the strength of these relationships varied. In contrast, SatisfactionScore showed no clear linear pattern, suggesting a limited direct association with customer spending.

3.3 Summary of Exploratory Findings

Summarize the main findings from the exploratory analysis and justify the modeling approach.

Key Findings

The exploratory analysis identified positive relationships between PurchaseAmount and Income, PromoUsed, PagesViewed, and WebsiteVisits, while SatisfactionScore showed no clear linear association. Since the response variable is continuous and the relationships with the main predictors are approximately linear, multiple linear regression was selected as the appropriate modeling approach. The candidate predictors will be further evaluated during model development.

4 Model Development

Develop and compare multiple linear regression models to identify the combination of predictors that best explains customer purchase amount.

4.1 Baseline Model

Develop a simple linear regression model using the strongest individual predictor identified during the exploratory analysis.

Model Equation

$$\text{PurchaseAmount} = \beta_0 + \beta_1 \times \text{Income}$$

```
# Fit the baseline regression model
baseline_model <- lm(
  PurchaseAmount ~ Income,
  data = marketing
)

# Display model summary
summary(baseline_model)

##
## Call:
## lm(formula = PurchaseAmount ~ Income, data = marketing)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -55.924 -12.137  -0.061  12.156  59.063
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  4.543e+01  9.069e-01   50.10  <2e-16 ***
## Income       4.940e-04  9.737e-06   50.73  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 17.43 on 4998 degrees of freedom
## Multiple R-squared:  0.3399, Adjusted R-squared:  0.3398
## F-statistic: 2574 on 1 and 4998 DF, p-value: < 2.2e-16

# Calculate confidence intervals
confint(baseline_model)

##              2.5 %      97.5 %
## (Intercept) 4.365389e+01 4.720955e+01
## Income      4.748656e-04 5.130421e-04

# Model selection criteria
AIC(baseline_model)

## [1] 42772.86

BIC(baseline_model)

## [1] 42792.42
```

Key Findings

Income was selected as the predictor for the baseline model because it showed the strongest linear relationship with PurchaseAmount during the exploratory analysis ($r = 0.58$). The model was statistically significant ($p < 0.001$) and explained approximately 34.0% of the variability in customer spending. Although this provides a useful starting point, the relatively modest explanatory power suggests that additional predictors may improve model performance.

4.2 Multiple Regression Model

Develop a multiple linear regression model including all candidate predictors identified during the exploratory analysis.

Model Equation

$$\text{PurchaseAmount} = \beta_0 + \beta_1 \times \text{Income} + \beta_2 \times \text{WebsiteVisits} + \beta_3 \times \text{PagesViewed} + \beta_4 \times \text{PromoUsed} + \beta_5 \times \text{SatisfactionScore}$$

```
# Fit the multiple regression model
multiple_model <- lm(
  PurchaseAmount ~ Income +
    WebsiteVisits +
    PagesViewed +
    PromoUsed +
    SatisfactionScore,
  data = marketing
)

# Display model summary
summary(multiple_model)
```

```
##
## Call:
## lm(formula = PurchaseAmount ~ Income + WebsiteVisits + PagesViewed +
##     PromoUsed + SatisfactionScore, data = marketing)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -41.464  -6.926   0.015   6.985  34.782
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.851e-01  8.482e-01   0.218   0.827
## Income         5.027e-04  5.667e-06  88.717 <2e-16 ***
## WebsiteVisits  2.002e+00  4.519e-02  44.287 <2e-16 ***
## PagesViewed    1.480e+00  2.620e-02  56.490 <2e-16 ***
## PromoUsed      2.001e+01  2.869e-01  69.750 <2e-16 ***
## SatisfactionScore -6.530e-02  1.249e-01  -0.523   0.601
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.13 on 4994 degrees of freedom
## Multiple R-squared:  0.7769, Adjusted R-squared:  0.7767
## F-statistic: 3479 on 5 and 4994 DF, p-value: < 2.2e-16
```

```
# Calculate confidence intervals
confint(multiple_model)
```

```
##              2.5 %      97.5 %
## (Intercept) -1.4777717262 1.847993e+00
## Income      0.0004916074 5.138252e-04
## WebsiteVisits 1.9129596758 2.090163e+00
## PagesViewed  1.4289309509 1.531676e+00
## PromoUsed    19.4477472569 2.057258e+01
## SatisfactionScore -0.3100775651 1.794855e-01
```

```
# Model selection criteria
AIC(multiple_model)
```

```
## [1] 37356.49
```

```
BIC(multiple_model)
```

```
## [1] 37402.11
```

Key Findings

The multiple regression model substantially improved the model performance compared with the baseline model. The explained variance increased from 34.0% to 77.7%, indicating that customer spending is better explained when demographic and behavioral variables are considered together. Among the candidate predictors, SatisfactionScore was the only variable that was not statistically significant ($p = 0.601$), making it the first candidate for removal during model refinement.

4.3 Refined Regression Model

Refine the multiple regression model by removing non-significant predictors while maintaining model performance.

Model Equation

$$\text{PurchaseAmount} = \beta_0 + \beta_1 \times \text{Income} + \beta_2 \times \text{WebsiteVisits} + \beta_3 \times \text{PagesViewed} + \beta_4 \times \text{PromoUsed}$$

```
# Fit the refined regression model
refined_model <- lm(
  PurchaseAmount ~ Income +
    WebsiteVisits +
    PagesViewed +
    PromoUsed,
  data = marketing
)
```

```
# Display model summary
summary(refined_model)
```

```
##
## Call:
## lm(formula = PurchaseAmount ~ Income + WebsiteVisits + PagesViewed +
##     PromoUsed, data = marketing)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -41.550  -6.915   0.003   6.964  34.700
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -1.386e-02  7.580e-01  -0.018   0.985
## Income        5.027e-04  5.666e-06  88.729  <2e-16 ***
## WebsiteVisits  2.002e+00  4.519e-02  44.295  <2e-16 ***
## PagesViewed    1.480e+00  2.620e-02  56.492  <2e-16 ***
## PromoUsed      2.001e+01  2.869e-01  69.757  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.13 on 4995 degrees of freedom
## Multiple R-squared:  0.7769, Adjusted R-squared:  0.7767
## F-statistic: 4349 on 4 and 4995 DF, p-value: < 2.2e-16
```

```
# Calculate confidence intervals
confint(refined_model)
```

```
##              2.5 %      97.5 %
## (Intercept) -1.499958476 1.472238e+00
## Income       0.000491629 5.138447e-04
## WebsiteVisits 1.913120288 2.090307e+00
## PagesViewed   1.428835151 1.531570e+00
## PromoUsed     19.448152202 2.057290e+01
```

```
# Model selection criteria
AIC(refined_model)
```

```
## [1] 37354.76
```

```
BIC(refined_model)
```

```
## [1] 37393.87
```

Key Findings

Removing SatisfactionScore simplified the regression model without reducing its explanatory power. The refined model maintained the same goodness of fit while achieving lower AIC and BIC values, indicating a more parsimonious model. Since all remaining predictors were statistically significant ($p < 0.001$), this model was selected as the final regression model.

4.4 Model Comparison

Compare the regression models using goodness-of-fit and model selection criteria to identify the preferred model.

Metric	Baseline	Multiple	Refined
Predictors	1	5	4
Multiple R ²	0.3399	0.7769	0.7769
Adjusted R ²	0.3398	0.7767	0.7767
Residual Std. Error	17.43	10.13	10.13
AIC	42772.86	37356.49	37354.76
BIC	42792.42	37402.11	37393.87

Key Findings

The baseline model established a useful benchmark, explaining 34.0% of the variation in PurchaseAmount using Income alone. Adding behavioral variables substantially improved the model, increasing the explained variance to 77.7%. Removing the non-significant predictor (SatisfactionScore) reduced model complexity while maintaining the same predictive performance and improving both AIC and BIC. Therefore, the refined regression model was selected as the final model for subsequent analysis.

5 Model Diagnostics

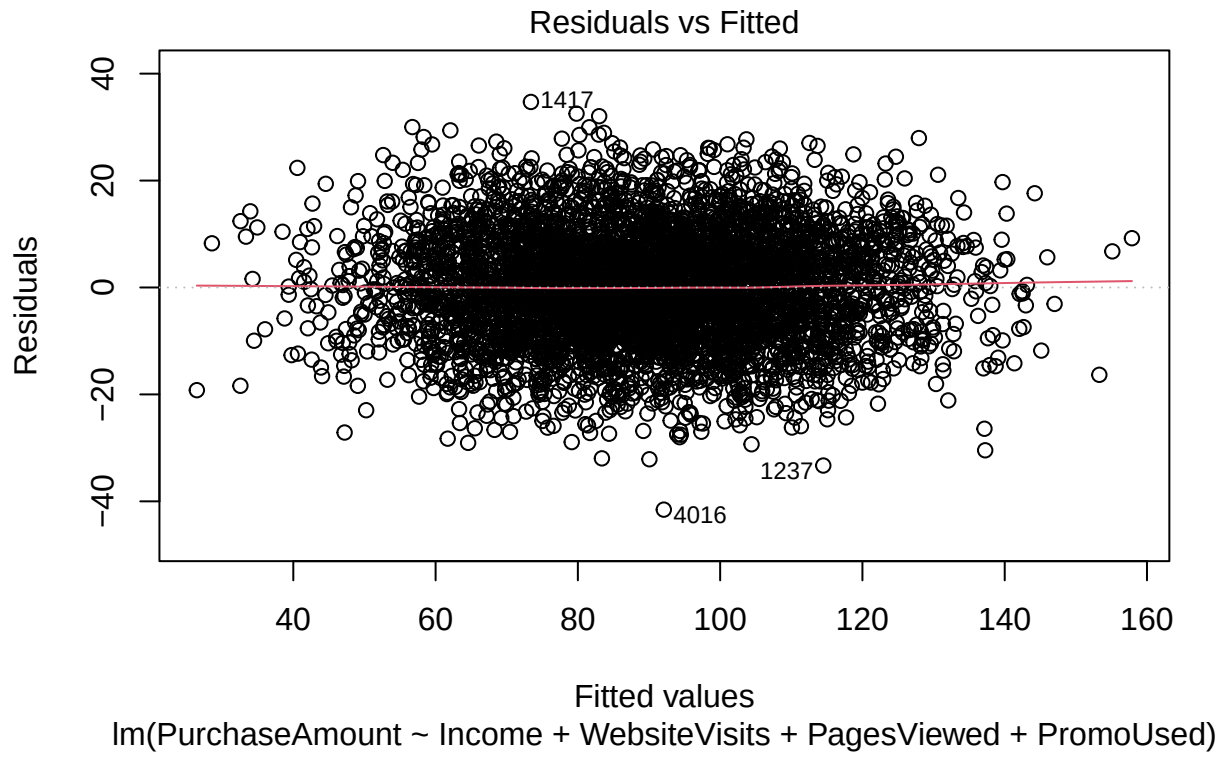
Evaluate whether the final regression model satisfies the key assumptions of multiple linear regression.

5.1 Linearity

Assess whether the relationship between the predictors and the response variable is approximately linear.

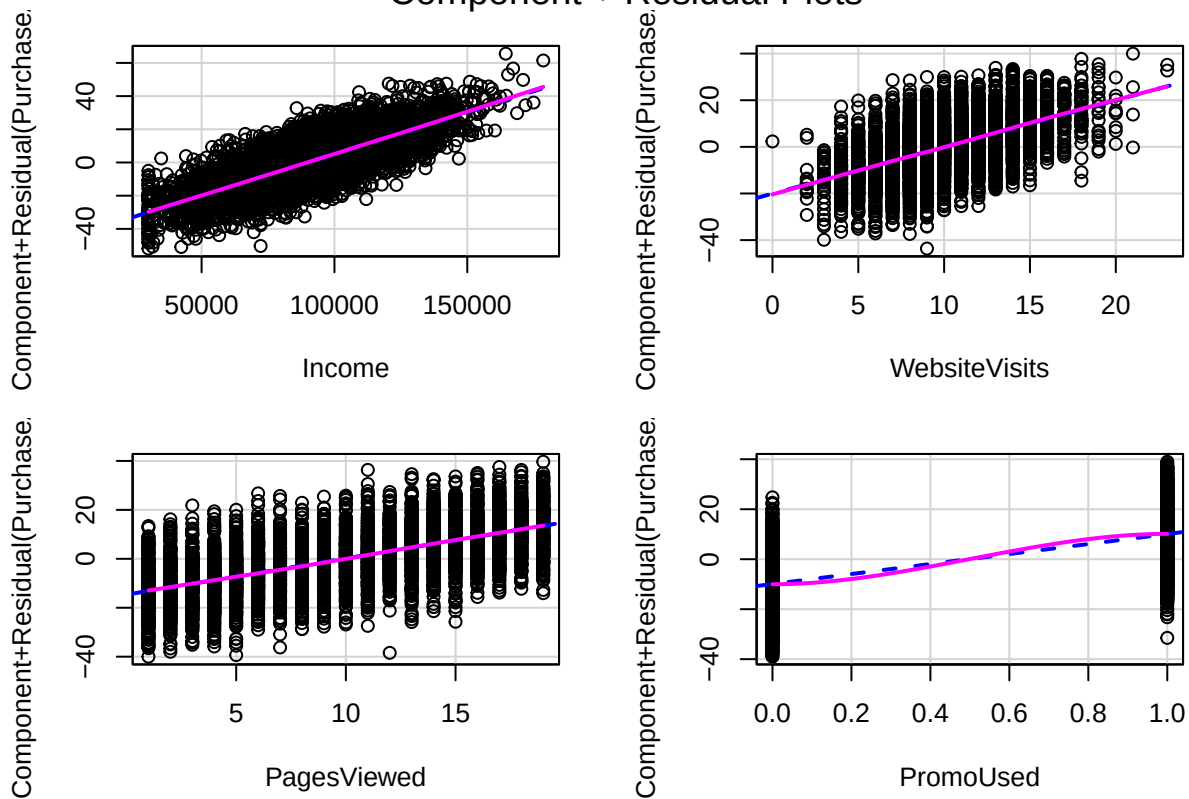
```
library(car)

# Residuals vs Fitted Values
plot(refined_model, which = 1)
```

```
# Component + Residual (Partial Residual) Plots  
crPlots(refined_model)
```

Component + Residual Plots



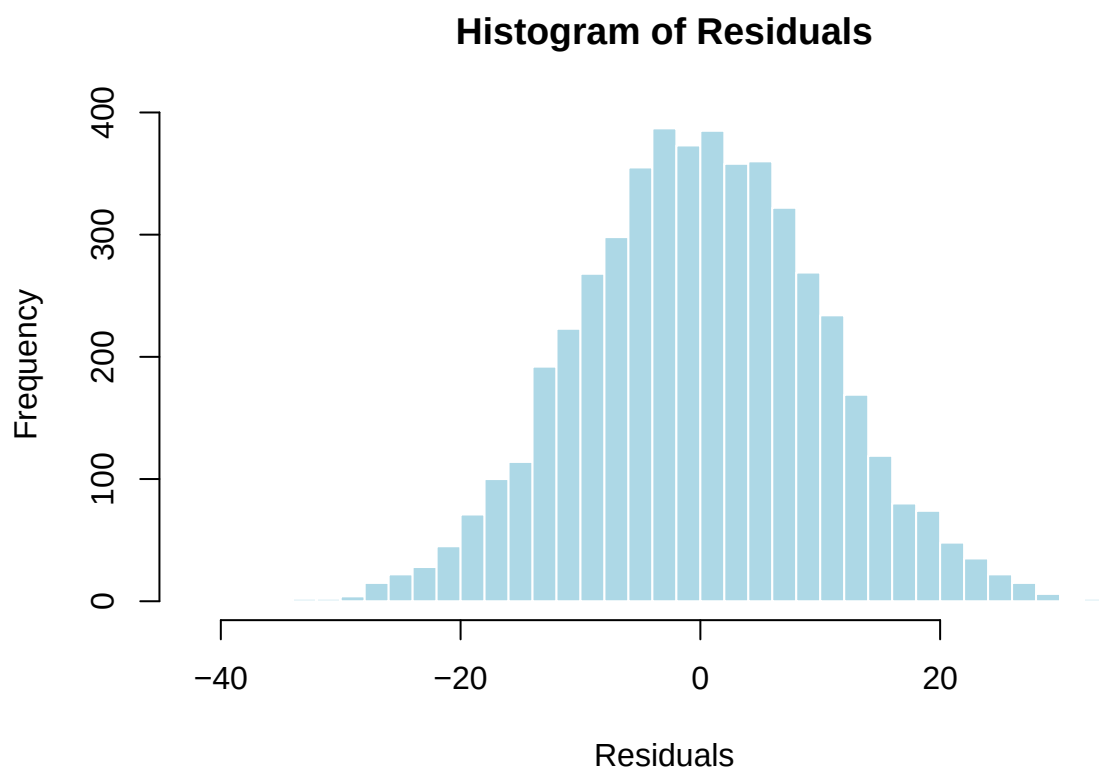
Key Findings

The Residuals vs Fitted plot showed residuals randomly scattered around zero with no systematic pattern, supporting the assumption of linearity for the model as a whole. The Component + Residual plots indicated approximately linear relationships for each retained predictor (Income, WebsiteVisits, PagesViewed, and PromoUsed). Therefore, no substantial evidence of non-linearity was observed, and no corrective action was required.

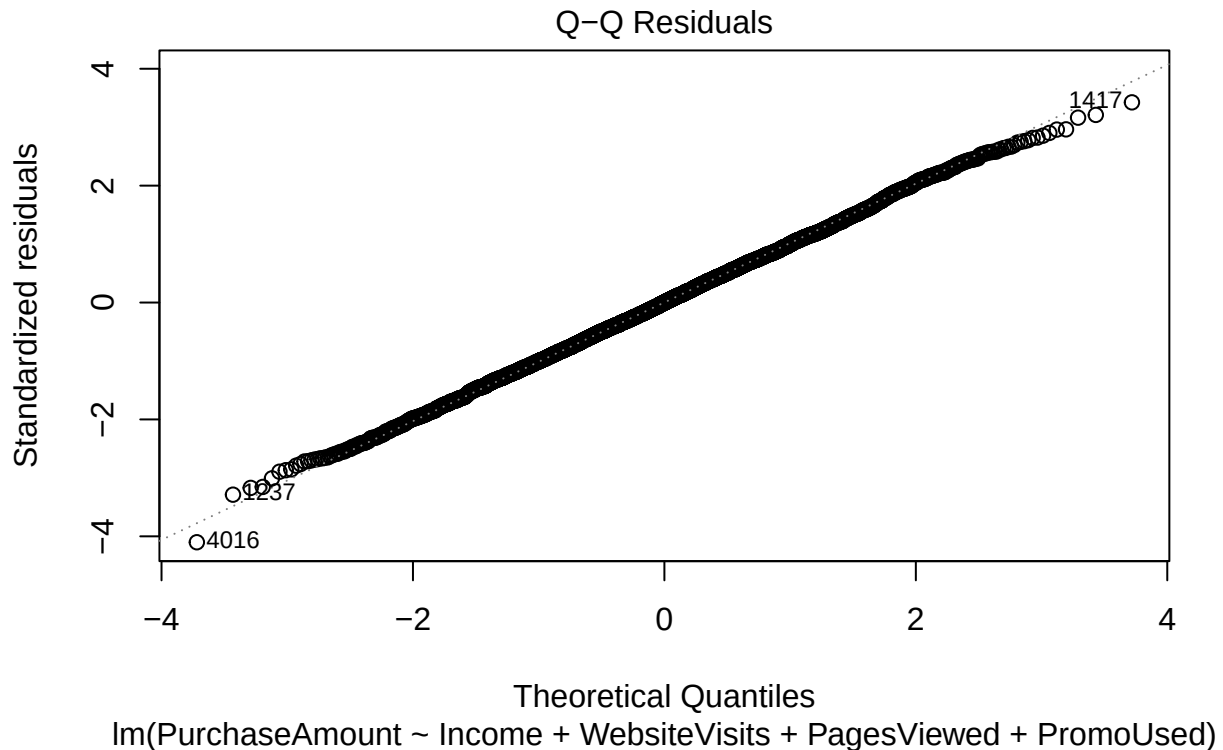
5.2 Normality of Residuals

Assess whether the residuals of the final regression model are approximately normally distributed.

```
# Histogram of residuals
hist(
  residuals(refined_model),
  breaks = 30,
  main = "Histogram of Residuals",
  xlab = "Residuals",
  col = "lightblue",
  border = "white"
)
```



```
# Q-Q Plot  
plot(refined_model, which = 2)
```



```
# Shapiro-Wilk Test
shapiro.test(residuals(refined_model))
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(refined_model)
## W = 0.9997, p-value = 0.7172
```

Key Findings

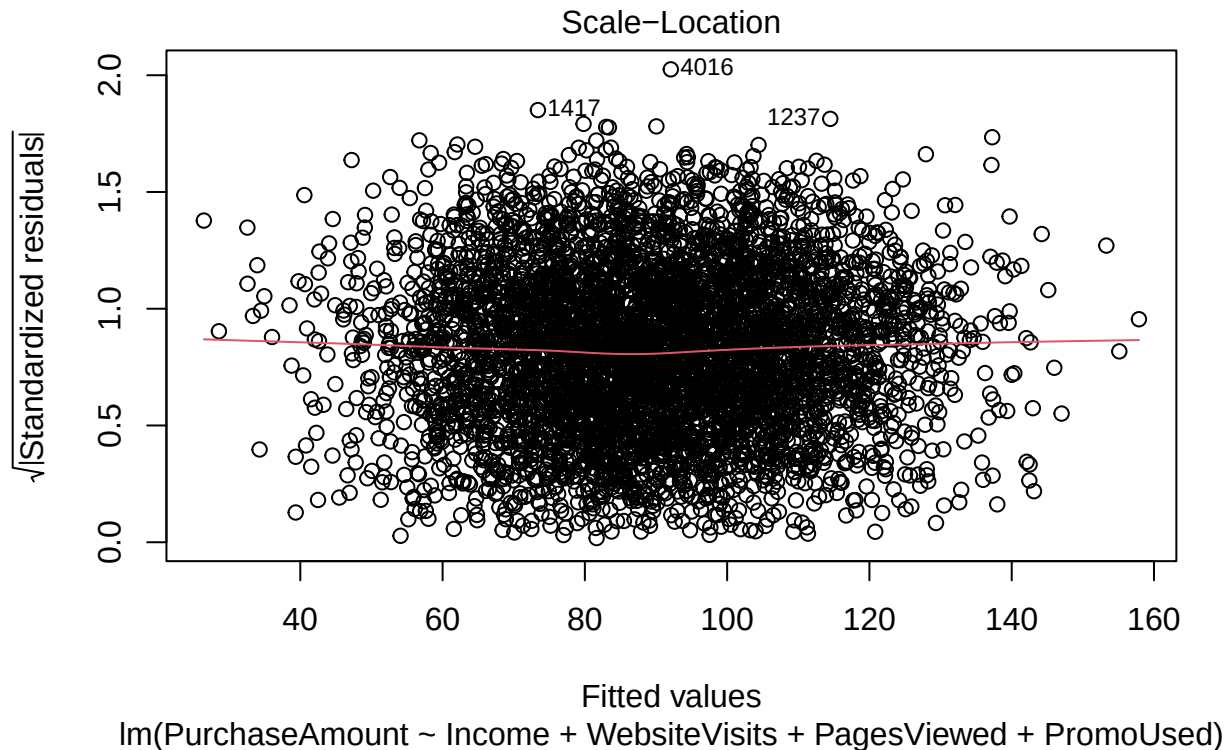
The histogram and Q-Q plot indicated that the residuals were approximately normally distributed, with only minor deviations in the tails. The Shapiro-Wilk test also failed to reject the null hypothesis of normality ($W = 0.9997$, $p = 0.7172$), providing additional statistical evidence that the residuals follow a normal distribution. Therefore, the normality assumption was considered satisfied, and no corrective action was required.

5.3 Homoscedasticity

Assess whether the residuals exhibit constant variance across the range of fitted values.

```
library(lmtest)

# Scale-Location Plot
plot(refined_model, which = 3)
```



```
# Breusch-Pagan Test
bptest(refined_model)
```

```
##
## studentized Breusch-Pagan test
##
## data: refined_model
## BP = 2.3799, df = 4, p-value = 0.6663
```

Key Findings

The Scale-Location plot showed a relatively constant spread of residuals across the fitted values, with no evidence of a funnel-shaped pattern. The Breusch-Pagan test also failed to detect heteroscedasticity (BP = 2.38, $p = 0.6663$). Together, these results indicate that the homoscedasticity assumption was satisfied, and no corrective action was required.

5.4 Multicollinearity

Assess whether the independent variables are highly correlated with one another, which could affect the stability of the regression coefficients.

```
library(car)

vif(refined_model)
```

##	Income	WebsiteVisits	PagesViewed	PromoUsed
##	1.001380	1.000246	1.000433	1.001660

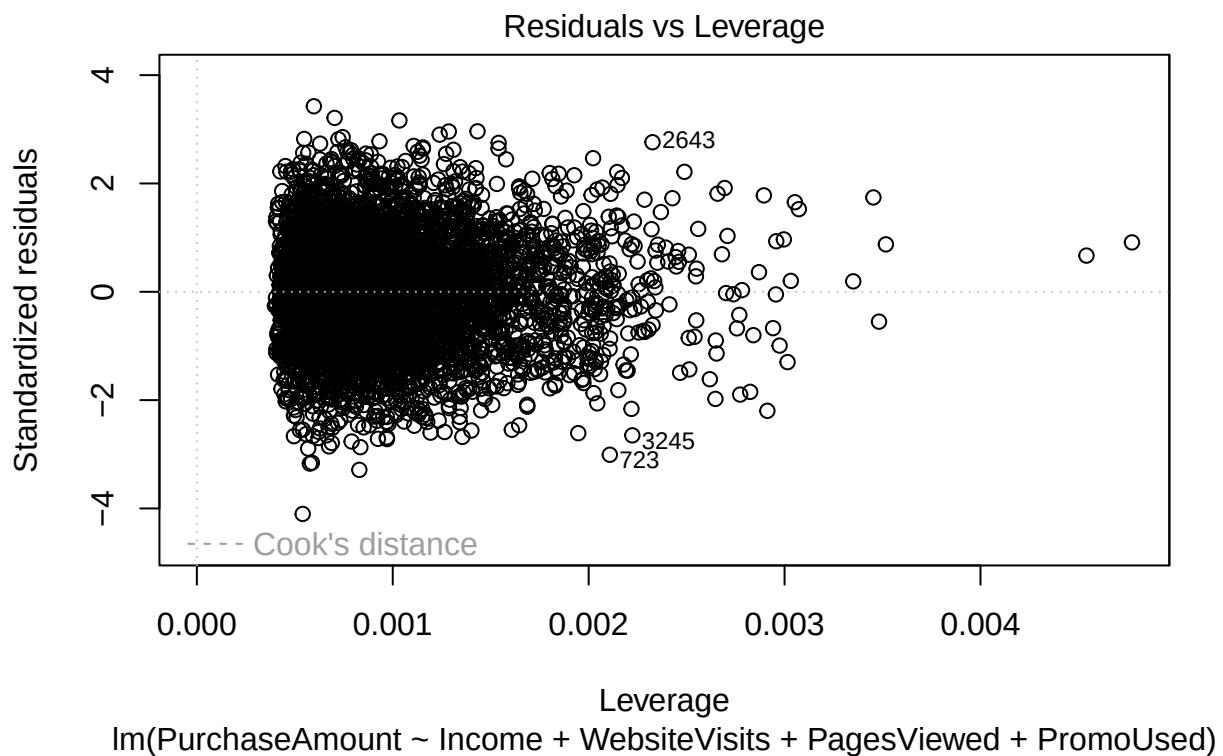
Key Findings

The Variance Inflation Factor (VIF) values for all retained predictors were approximately equal to 1, indicating virtually no multicollinearity. This suggests that each predictor contributes unique information to the model, resulting in stable coefficient estimates and reliable statistical inference. Therefore, no corrective action was required.

5.5 Influential Observations (Cook's Distance)

Identify observations that exert an unusually large influence on the estimated regression model.

```
plot(refined_model, which = 5)
```



```
cook <- cooks.distance(refined_model)
threshold <- 4 / nrow(marketing)
cat("Cook's Distance threshold:", round(threshold, 5), "\n")
```

```
## Cook's Distance threshold: 8e-04
```

```
cat("Maximum Cook's Distance:", round(max(cook), 5), "\n")
```

```
## Maximum Cook's Distance: 0.00382
```

```
cat("Potentially influential observations:", sum(cook > threshold), "\n")
```

```
## Potentially influential observations: 255
```

Key Findings

The Cook's Distance analysis identified 255 observations above the conservative screening threshold (4/n). However, the largest Cook's Distance was only 0.0038, indicating that no individual observation exerted excessive influence on the regression model. Therefore, all observations were retained, and no corrective action was required.

5.6 Diagnostic Summary

Summarize the results of the regression diagnostic tests and assess whether the final model satisfies the assumptions of multiple linear regression.

Assumption	Diagnostic Method	Result	Action Required
Linearity	Residuals vs. Fitted & Component + Residual Plots	Satisfied	No
Normality	Histogram, Q-Q Plot & Shapiro-Wilk Test	Satisfied	No
Homoscedasticity	Scale-Location Plot & Breusch-Pagan Test	Satisfied	No
Multicollinearity	Variance Inflation Factor (VIF)	Not detected	No
Influential Observations	Cook's Distance	No excessive influence detected	No

Key Findings

The final regression model satisfied all major assumptions of multiple linear regression. Diagnostic plots and statistical tests supported the adequacy of the model, indicating linear relationships, approximately normal residuals, constant variance, no evidence of multicollinearity, and no excessively influential observations. The refined regression model was therefore retained as the final model for interpretation.

6 Conclusion

This notebook documented the complete regression modeling process, including exploratory data analysis, model development, model comparison, and diagnostic assessment. Based on statistical performance and diagnostic validation, the refined multiple linear regression model was selected as the final model. The interpretation of the regression coefficients, business implications, limitations, and recommendations are presented in the accompanying report.

Next Step

The accompanying report builds upon this technical analysis by interpreting the final regression model in a marketing context, discussing business implications, limitations, and opportunities for future improvement.